PARKING LTD.

CAPSTONE MANUAL

WALKTHROUGH GUIDE

Demonstration runbook for recorded simulations and live VIGI gates.

Automated Parking Management System

Group code: CAP-2520-IT

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For capstone presenters, operators, and reviewers conducting a guided demonstration.

1. Demonstration map

Use this guide to show the complete vehicle lifecycle: entry recognition, QR issuance, parking assignment, business-rule handling, cashier or tenant action, exit authorization, and the records visible in the main frontend. The recorded and live-gate modes share the PHP API and SQLite application database.

Mode	Entrance	Sectional feed	Exit
Recorded run	Recorded entrance clip	Selected zone snapshot, parking clip, and departure clip	Recorded exit clip
Live gates	VIGI NVR channel 1	Optional recorded sectional feed in combined mode	VIGI NVR channel 13
Direct live_vigi.py	VIGI NVR channel 1	No sectional recognition in this command	VIGI NVR channel 13

Important operational boundary

The application records clearance and session state. It does not actuate a physical boom barrier or read radar loops in the current code. A guard operates the physical barrier and verifies that the vehicle has actually passed.

2. Before the demo

- Use the project root and start the PHP app on 127.0.0.1:8080. Start it with LOCAL_PAYMENTS_ENABLED=1 when demonstrating reservation checkout. Local simulated payments move no money.
- Sign in to the four role views: Driver, Admin, Cashier, and Joey's tenant Vendor. Confirm that the running app and computer vision tools use backend/database/parking.db.
- For live feeds, connect the operator computer to the NVR network, confirm reachability of 192.168.31.2 on RTSP port 554, and load the credentials into environment variables. Never show or paste the password in a screenshot or command history.
- Confirm the entrance and exit views, model files, clips, bay calibration JSON, and camera event token are available. Run --detection-only before a live business-flow demonstration.
- Choose a fresh plate/session and clear any old open session before starting. Tell the audience which mode is being shown and that the database is the live demo database.

```
LOCAL_PAYMENTS_ENABLED=1 php -S 127.0.0.1:8080 backend/router.php
source .venv-cv/bin/activate
python computer_vision/live_vigi.py --detection-only
```

3. Recorded business-rule demonstration

The recorded runner follows the same entry, cashier, tenant, API, and exit paths as the live system. It uses actual camera clips for gates, and recorded sectional clips for occupancy. For each zone run: snapshot -> entrance -> assignment -> parking confirmation -> rule action -> bay departure -> exit. Refresh the frontend to show database changes.

Zone	Recorded target bay	Entrance / exit car	Snapshot
A	A-06	XPANDER	ZONE A SNAPSHOT.mp4
B	B-05	XPANDER	ZONE B SNAPSHOT.mp4
C	C-01	XPANDER	ZONE C SNAPSHOT.mp4
D	D-02	FD	ZONE D SNAPSHOT.mp4
E	E-01	FD	ZONE E SNAPSHOT.mp4

The Zone D clip filenames say D-01, but the current profile maps the recorded FD vehicle to D-02. Confirm the snapshot recognizes the selected target as vacant before entry. In Zone A, vehicles already in the snapshot are assumed VIP/PWD authorized for simulation. A newly entering non-VIP car is assigned to the nearest available eligible zone, Zone B first, even when its recorded parking clip shows it in A-06.

Recorded clip catalog

Zone	Entrance and exit gate clips	Snapshot, parking, and departure clips
A	ENTRANCE_XPANDER.mov EXIT_XPANDER.mov	ZONE A SNAPSHOT.mp4 PARK_XPANDER-ZONEA-06v1.mp4 EXITPARK_XPANDER-ZONEA-06v1.mp4
B	ENTRANCE_XPANDER.mov EXIT_XPANDER.mov	ZONE B SNAPSHOT.mp4 PARK_XPANDER-ZONEB-05V1.mp4 EXITPARK_XPANDER-ZONEB-05v1.mp4
C	ENTRANCE_XPANDER.mov EXIT_XPANDER.mov	ZONE C SNAPSHOT.mp4 PARK_XPANDER-ZONEC-01v1.mp4 EXITPark_XPANDER-ZONEC-01v1.mp4
D	ENTRANCE_FD.mov EXIT_FD.mov	ZONE D SNAPSHOT.mp4 PARK FD ZONE D-01.mp4 EXIT FD ZONE D-01.mp4
E	ENTRANCE_FD.mov EXIT_FD.mov	ZONE E SNAPSHOT.mp4 PARK FD ZONE E-01.mp4 EXIT FD ZONE E-01.mp4

Zone C has the only recorded roadway-enforcement feed: VIOLATION_XPANDER-ZONECv1.mp4. The enforcement business-rule run uses this clip and skips the bay-departure clip before exit.

4. Demo one complete recorded run

1. **Start the app and activate the CV environment**

   Run the PHP server in one terminal and activate .venv-cv in another. Keep both running while the app and simulation operate.

2. **Choose a business-rule command**

   For example: python computer_vision/recorded_test_cases/grace.py --zone A. The per-rule scripts include grace, normal_fare, reservation, reserved_slot, tenant, and enforcement. Enforcement has a recorded demonstration only for Zone C.

3. **Review the pre-entry snapshot**

   The sectional window displays all calibrated bays and detected occupancy. The snapshot updates zone occupancy in the application database. The selected recorded target must be observed vacant. Pause and review if it is unknown or occupied.

4. **Show entry recognition and token issuance**

   The entrance clip runs through the shared PlateDetector, which confirms repeated plate reads and samples entrance color. The backend creates a parked session, allocates a bay, records a timestamp, and issues an entry QR even when a plate is readable. The terminal prints the QR and token.

5 **Show the parking clip**

The sectional detector analyzes bays with the vehicle detector and calibrated polygons. A continuous 10-second target-bay occupancy confirms parking. Plate OCR can add identity evidence when readable; clip labels are scenario labels, not camera OCR.

6 **Apply the business rule in the matching UI**

Use Cashier POS for grace, normal fare, reservation balance, lost ticket, and violation fees. Use Tenant Validator for Joey's 20-minute window and ₱30 base-fee subsidy. Use Admin > Violations Log for the staff decision on alerts.

7 **Show departure and exit clearance**

For ordinary cases, play the bay-departure clip and then the exit clip. The exit detector must match the active plate and satisfy payment, tenant-window, color-review, and alert-review requirements. If denied, complete the required action and replay the exit stage.

8 **Verify closure**

Refresh Cashier, Admin, and Dashboard. Show EXITED session state, the released assigned slot, zone count, payment or fine status, and audit record. A physical guard still controls the barrier.

5. Business rules to demonstrate

Business rule	Command pattern
Grace bypass	python computer_vision/recorded_test_cases/grace.py --zone A
Normal fare	python computer_vision/recorded_test_cases/normal_fare.py --zone A
Paid reservation	python computer_vision/recorded_test_cases/reservation.py --zone A
Another vehicle reserved bay	python computer_vision/recorded_test_cases/reserved_slot.py --zone A
Tenant validation	python computer_vision/recorded_test_cases/tenant.py --zone A
Driveway/double-parking enforcement	python computer_vision/recorded_test_cases/enforcement.py --zone C

For the first five rules, replace --zone A with B, C, D, or E to use that zone profile. Reservations must be made manually in the frontend for the matching plate and bay before reservation.py. The reserved-slot case requires another vehicle to hold the recorded bay. Complete each session before the next run.

Case	Expected result	Where to act
Grace turnaround	Up to 15 minutes from entry: ₱0 bypass (BYPASSED).	Cashier POS
Normal daytime stay	After grace: ₱30 flat fare. Unpaid exit is denied.	Cashier POS, then exit
Overnight stay	₱100 total overnight charge when the local stay crosses date or reaches 23:00; not ₱30 plus ₱100.	Cashier POS
Paid reservation	Driver reserves a specific bay in the frontend; demo checkout confirms it. Matching entry fulfills the booking.	Reserve Slot, Cashier/exit
Reserved-bay intrusion	Vehicle is allocated elsewhere, while sectional clip shows reserved-bay entry; staff must review before exit.	Admin > Violations Log
Tenant validation	Joey's vendor validation waives ₱30 and starts a 20-minute exit window. After expiry, the base fare is due.	Tenant Validator, Cashier

Case	Expected result	Where to act
Zone A priority	Zone A is VIP/PWD priority. Non-VIP new entrant gets an eligible bay elsewhere. A recorded clip showing unauthorized Zone A parking raises an alert after occupancy confirms.	Admin > Violations Log
Driveway/double parking	Stationary vehicle in a calibrated restricted region creates an evidence-backed alert. Recorded Zone C demo uses 20 seconds; regular monitor default is 20 seconds in this build.	Admin review, Cashier for confirmed fee
Unpaid violation	All ordinary confirmed penalties are ₱200. Rejected alert adds nothing. Lost-ticket charge is ₱100.	Cashier POS
Unreadable exit plate	Use the issued entry QR, check the session and supporting entrance color, settle all charges, then record QR exit. Color never authorizes exit.	Cashier POS

6. Live VIGI demonstration

There are two live presentations. Direct live_vigi.py opens channel 1 and channel 13 and publishes gate events; it does not run sectional occupancy. full_simulation.py live-gates adds a recorded sectional snapshot and parking replay between live gates. Both use the main database and app.

```
export VIGI_NVR_IP="192.168.31.2"
export VIGI_NVR_USERNAME="admin"
read -s 'VIGI_NVR_PASSWORD?NVR password: '
export VIGI_NVR_PASSWORD
echo
python computer_vision/live_vigi.py
```

For an unreadable live entrance, press u with exactly one vehicle visible to create a QR-only walk-in. Press r with exactly one vehicle visible to scan an active paid reservation QR. Both methods record the observed entrance color; the issued QR is the exit fallback. Detection-only mode is safe for feed checks and makes no API changes.

7. Presentation close and reset

- Show the Admin audit trail, session logs, ALPR event, alert evidence, cashier transaction, and final map state. This is evidence of what the software recorded; it does not prove physical barrier actuation.
- Do not run the next clip using a plate with an open session. Finish cashier/tenant actions and exit, or use Admin > Simulation Sessions recovery with a reason after resolving charges.
- Do not delete the main database to reset a demo. Snapshot observations and simulation session events remain for the audit trail; manage live occupied slots through the normal exit or admin recovery process.